Practice workbook · Proving identities with double-angle and addition formulae

Rewriting a longer trig expression into a given form by substituting double-angle and addition results, using sin^2 + cos^2 = 1 and tan = sin/cos.

14 question parts: 1 worked-example retry and 13 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-23

1. Question 9 · part b · 5 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part b only.

9 (a) Using the formulae on page 2, show that

(i) $\cos^2 A = \dfrac{\cos 2A + 1}{2}$

(ii) $\sin^2 A = \dfrac{1 - \cos 2A}{2}$

(4)

(b) Show that

$$(2\sin x - \cos x)(\sin x - 3\cos x) = \frac{1}{2}(\cos 2x - 7\sin 2x + 5)$$

(5)

$$y = (2\sin x - \cos x)(\sin x - 3\cos x)$$

(c) Solve, for $0° \leqslant x \leqslant 180°$ the equation, $\dfrac{dy}{dx} = 0$

Give your answers to the nearest whole number.

(4)

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1. Question 9 · part b · 5 marks · Working and answer

Working and answer

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2. Question 10 · part b · 4 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

10 (a) Use the formula for $\cos(A + B)$ to show that $\cos 2A = 2\cos^2 A - 1$

(2)

(b) Show that $\cos 4A = 8\cos^4 A - 8\cos^2 A + 1$

(4)

(c) Solve the equation $\cos^2\left(\frac{\theta}{4} + \frac{\pi}{24}\right)\left[\cos^2\left(\frac{\theta}{4} + \frac{\pi}{24}\right) - 1\right] = -\frac{1}{16} \qquad 0 \leqslant \theta < 2\pi$

Give your answers in terms of π.

(5)

$$f(A) = 4\cos^4 A - 4\cos^2 A + 1$$

(d) Using calculus, find the exact value of $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} f(A)\,dA$

Give your answer in the form $a\pi - b\sqrt{c}$ where a and b are fractions in their lowest terms and c is a prime number.

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2. Question 10 · part b · 4 marks · Working and answer

Working and answer

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3. Question 7 · part a · 4 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

7 Given that

$$\frac{3+\sin^2\theta}{\cos\theta-2}=3\cos\theta$$

(a) show that $\cos\theta=-\frac{1}{2}$

(4)

(b) Hence solve the equation

$$\frac{3+\sin^2 3x}{\cos 3x-2}=3\cos 3x \quad \text{for } 0^\circ \leqslant x < 180^\circ$$

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P 6 6 3 0 7 A 0 1 8 3 2

3. Question 7 · part a · 4 marks · Working and answer

Working and answer

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4. Question 11 · part a · 3 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

11 $f(\theta) = (2\cos\theta - \sin\theta)(2\sin\theta + \cos\theta)$

(a) Show that $f(\theta) = \frac{3}{2}\sin 2\theta + 2\cos 2\theta$

(3)

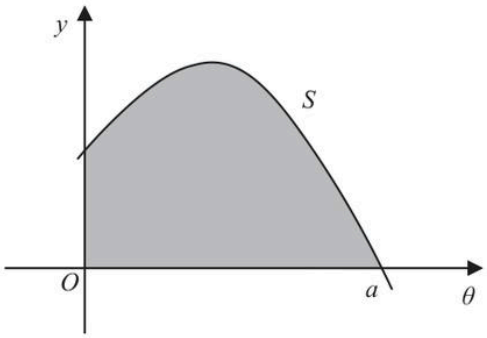


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows part of the curve S with equation $y = f(\theta) + 2$

Given that S intersects with the θ-axis at the point with coordinates $(a, 0)$

(b) using $\sin^2\theta + \cos^2\theta = 1$, or otherwise, show that $a = \frac{\pi}{2}$

(5)

(c) Using algebraic integration, find the exact area bounded by S, the positive θ-axis and the positive y-axis shown shaded in Figure 4

(3)

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4. Question 11 · part a · 3 marks · Working and answer

Working and answer

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5. Question 10 · part b · 4 marks

2023 JAN · Paper 02

Further practice · Attempt part b only.

10 Using formulae given on page 2

(a) show that

(i) $\sin 2\theta = 2\sin\theta\cos\theta$

(ii) $\cos 2\theta = 2\cos^2\theta - 1$

(5)

Given that $\theta \neq (90° + 180°n)$ where $n \in \mathbb{Z}$

(b) use the results from part (a) to show that $\sin 2\theta - \tan\theta$ can be written as $\tan\theta\cos 2\theta$

(4)

(c) Solve for $0 < x < 360$

$$\sin 2x° - \tan x° = 0$$

(4)

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5. Question 10 · part b · 4 marks · Working and answer

Working and answer

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6. Question 11 · part b · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part b only.

11 Using formulae from page 2, show that

(a) (i) $\cos 2A = 2\cos^2 A - 1$ (3)

(ii) $\sin 2A = 2\sin A \cos A$ (1)

(b) Show that $\cos^3 A = \dfrac{\cos 3A + 3\cos A}{4}$ (4)

Hence, or otherwise,

(c) solve, giving exact values in terms of π

$$8\cos^3\left(\frac{\theta}{2}\right) - 6\cos\left(\frac{\theta}{2}\right) - 1 = 0 \qquad \text{for } 0 \leqslant \theta \leqslant 2\pi$$ (4)

(d) use algebraic integration to find the exact value of

$$\int_0^{\frac{\pi}{6}} \left(4\cos^3\theta - \sin 2\theta\right) \mathrm{d}\theta$$ (4)

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6. Question 11 · part b · 4 marks · Working and answer

Working and answer

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7. Question 8 · part b · 3 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

8 (a) Write down the value of k such that $\sin 2A = k \sin A \cos A$

(1)

$$g(A) = 2 + 3\cos A - \sin A - 3\sin 2A - 2\cos^2 A$$

Given that $g(A)$ can be written in the form $(p\cos A - \sin A)(q - r\sin A)$ where p, q and r are integers,

(b) find the value of p, the value of q and the value of r.

(3)

(c) Hence solve, in radians to 3 significant figures where appropriate, the equation

$$g(2\theta) = 0 \quad \text{for} \quad 0 \leqslant \theta < \pi$$

(6)

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7. Question 8 · part b · 3 marks · Working and answer

Working and answer

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8. Question 1 · part b · 3 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

1 (a) Use the formula for $\cos(A + B)$ to show that $\cos 2A = 1 - 2\sin^2 A$

(2)

Given that $\sin A = \frac{x+1}{2}$ and $\cos 2A = \frac{4-y}{3}$

(b) show that $y = \frac{1}{2}(3x^2 + 6x + 5)$

(3)

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8. Question 1 · part b · 3 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

Question 1 continued

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(Total for Question 1 is 5 marks)

8. Question 1 · part b · 3 marks · Working and answer

Working and answer

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9. Question 11 · part b · 5 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

11 (a) Use a formula on page 2 to show that $\sin^2 A = \frac{1}{2}(1 - \cos 2A)$ (3)

(b) Show that $\sin^4 x + \cos^4 x = \frac{3 + \cos 4x}{4}$ (5)

(c) Hence solve, in degrees to one decimal place, the equation

$$8\sin^4\left(\frac{\theta}{2}\right) + 8\cos^4\left(\frac{\theta}{2}\right) = 5\sin(2\theta) + 6 \quad \text{for} \quad 0^\circ \leqslant \theta < 180^\circ$$

(4)

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9. Question 11 · part b · 5 marks · Working and answer

Working and answer

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10. Question 10 · part c · 4 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

10 (a) Use the factor theorem to show that $(4x - 3)$ is a factor of

$$16x^3 + 11x - 15$$

(2)

(b) Using formulae given on page 2, show that

(i) $\sin 2\theta = 2\sin\theta\cos\theta$

(ii) $\cos 2\theta = 2\cos^2\theta - 1$

(5)

(c) Show that the equation

$$27\cos\theta\cos 2\theta + 19\sin\theta\sin 2\theta - 15 = 0$$

becomes the equation

$$16x^3 + 11x - 15 = 0$$

by using the substitution $x = \cos\theta$

(4)

(d) Hence show that any solution of the equation

$$27\cos\theta\cos 2\theta + 19\sin\theta\sin 2\theta - 15 = 0$$

is given by $\cos\theta = \frac{3}{4}$

(4)

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10. Question 10 · part c · 4 marks · Working and answer

Working and answer

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11. Question 10 · part b · 4 marks

2023 NOV · Paper 01

Further practice · Attempt part b only.

10 (a) Using formulae on page 2, show that

(i) $\sin 2A = 2\sin A\cos A$

(ii) $\cos 2A = 2\cos^2 A - 1$

(3)

$$f(\theta) = \frac{2\tan\theta}{1+\tan^2\theta}$$

(b) Show that $f(\theta) = \sin 2\theta$

(4)

(c) Solve, in radians to 3 significant figures, for $-\frac{\pi}{2} \leqslant x \leqslant \frac{\pi}{2}$, the equation

$$5\tan\left(x+\frac{\pi}{6}\right) = \left[1+\tan^2\left(x+\frac{\pi}{6}\right)\right]\left[1-2\cos^2\left(x+\frac{\pi}{6}\right)\right]$$

(6)

(d) Using calculus, find the exact value of

$$\int_0^{\frac{\pi}{2}} \left(\frac{4\tan\theta}{1+\tan^2\theta} - \cos 5\theta + 2\right) d\theta$$

(4)

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11. Question 10 · part b · 4 marks · Working and answer

Working and answer

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12. Question 11 · part a · 4 marks

2025 JUN · Paper 02

Further practice · Attempt part a only.

11 (a) Show that $\cos^4\theta - \sin^4\theta = \cos 2\theta$

(4)

(b) Hence, or otherwise, solve the equation

$$8\cos^2\left(2\theta + \frac{\pi}{4}\right) - 3 = 2\cos^4\left(\theta + \frac{\pi}{8}\right) - 2\sin^4\left(\theta + \frac{\pi}{8}\right) \quad \text{for} \quad 0 \leqslant \theta < \pi$$

Give your solutions to 2 decimal places.

(7)

(c) Using calculus, find the exact value of $\int_{\frac{\pi}{16}}^{\frac{\pi}{8}} \left(\cos^4 2x - \sin^4 2x - 8\sin 4x\right) \mathrm{d}x$

Give your answer in the form $a - b\sqrt{2}$ where a and b are rational numbers.

(4)

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12. Question 11 · part a · 4 marks · Working and answer

Working and answer

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13. Question 11 · part b · 3 marks

2020 JAN · Paper 02

Further practice · Attempt part b only.

11 (a) Express the equation

$$3\sin(A - B) = \sin(A + B)$$

in the form $\tan A = k \tan B$, giving the value of the integer k.

(4)

(b) Given that $\theta \neq \frac{(2n+1)\pi}{2}$ where $n \in \mathbb{Z}$,

show that $\frac{\cos^4\theta - \sin^4\theta}{\cos^2\theta} = 1 - \tan^2\theta$

(3)

(c) Using the exact values of $\sin x°$, $\cos x°$ and $\tan x°$ for $x = 30, 45, 60$

show that

(i) $\cos 15° = \frac{\sqrt{6} + \sqrt{2}}{4}$

(2)

(ii) $\tan 255° = \frac{3 + \sqrt{3}}{3 - \sqrt{3}}$

(4)

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13. Question 11 · part b · 3 marks · Working and answer

Working and answer

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14. Question 11 · part b · 3 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

11 (a) Using a formula on page 2 show that $\cos 2A = 2\cos^2 A - 1$

(2)

(b) Hence show that $\left(2\cos^2 A - 1\right)^2 = \dfrac{\cos 4A + 1}{2}$

(3)

The curve with equation $y = \dfrac{\sin 2x}{2} + \dfrac{\left(2\cos^2 x - 1\right)^2}{2} + \dfrac{1}{8}$ has a stationary point P in the range $0 \leqslant x \leqslant \dfrac{\pi}{6}$

(c) Find the exact coordinates of P

(7)

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14. Question 11 · part b · 3 marks · Working and answer

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